

Initiating Search

November 5, 2025, 5:25 PM

• Search:

CAS Lexicon Search:

Mitsunobu reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (2,331)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,114)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	

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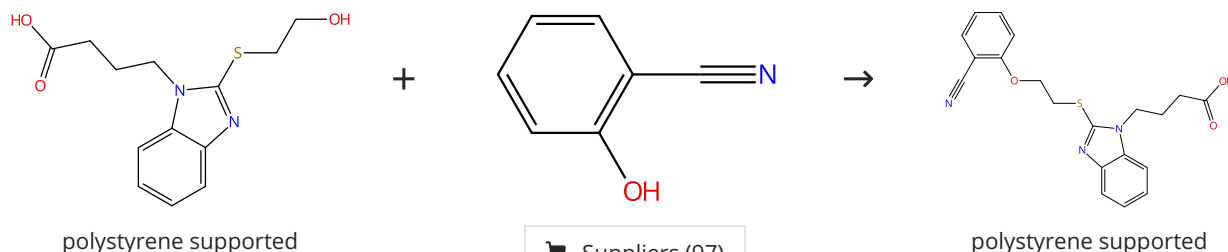
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (97)

31-012-CAS-16504589

Steps: 1 Yield: 100%

Solid phase syntheses of *S,N*-substituted 2-mercaptobenzimidazoles

1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Chloroform; rt; 0 °C; 3 h, rt

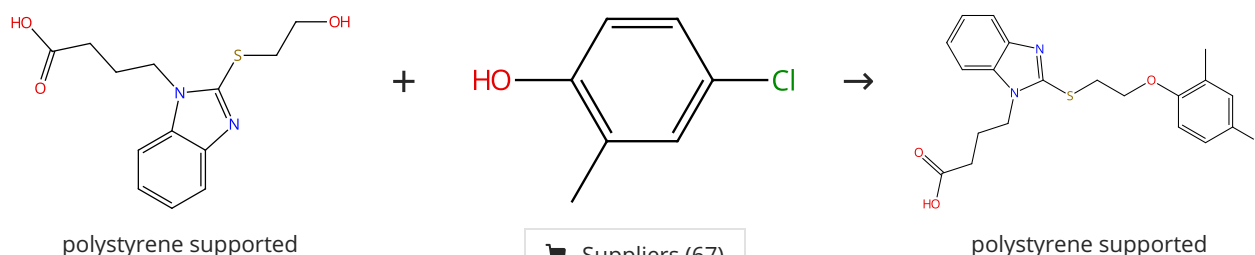
By: Susanto, Steven; et al

Experimental Protocols

RSC Advances (2016), 6(46), 39573-39576.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (67)

31-012-CAS-16504591

Steps: 1 Yield: 100%

Solid phase syntheses of *S,N*-substituted 2-mercaptobenzimidazoles

1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Chloroform; rt; 0 °C; 3 h, rt

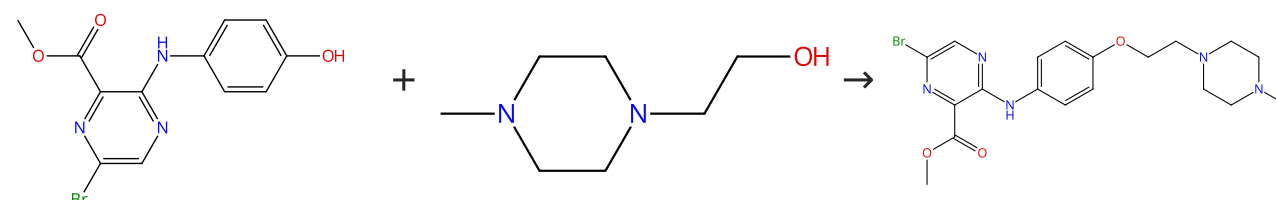
By: Susanto, Steven; et al

Experimental Protocols

RSC Advances (2016), 6(46), 39573-39576.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Supplier (1)

 Suppliers (80)

31-012-CAS-14408937

Steps: 1 Yield: 100%

Efficient Three-Step One-Pot Synthesis of a Novel 2,3,5-Substituted Pyrazine Library

1.1 Reagents: Triphenylphosphine (resin-bound), Di-*tert*-butyl azodicarboxylate
Solvents: Dichloromethane; 1 h, 0 °C; 16 h, rt

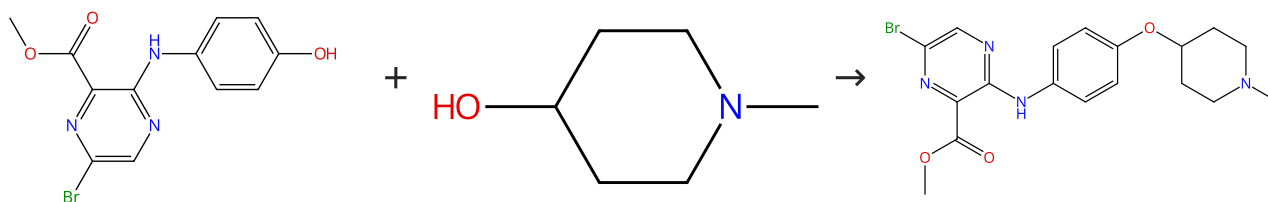
By: Delvare, Christian; et al

Experimental Protocols

ACS Combinatorial Science (2011), 13(5), 449-452.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (91)

31-012-CAS-12279997

Steps: 1 Yield: 100%

Efficient Three-Step One-Pot Synthesis of a Novel 2,3,5-Substituted Pyrazine Library

By: Delvare, Christian; et al

ACS Combinatorial Science (2011), 13(5), 449-452.

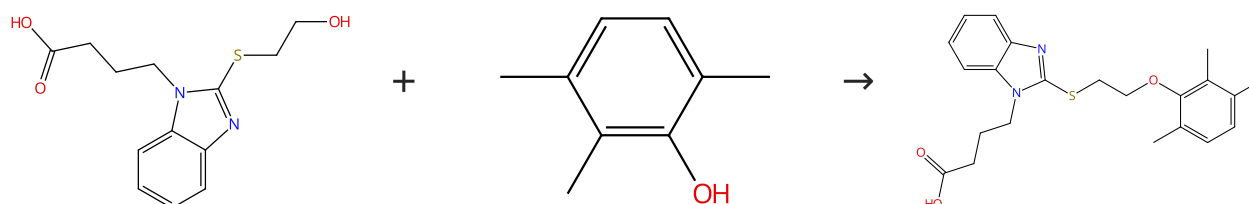
1.1 Reagents: Triphenylphosphine (resin-bound), Di-*tert*-butyl azodicarboxylate

Solvents: Dichloromethane; 1 h, 0 °C; 16 h, rt

Experimental Protocols

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



polystyrene supported

Suppliers (66)

polystyrene supported

31-012-CAS-16504592

Steps: 1 Yield: 100%

Solid phase syntheses of S,N-substituted 2-mercaptobenzimidazoles

By: Susanto, Steven; et al

RSC Advances (2016), 6(46), 39573-39576.

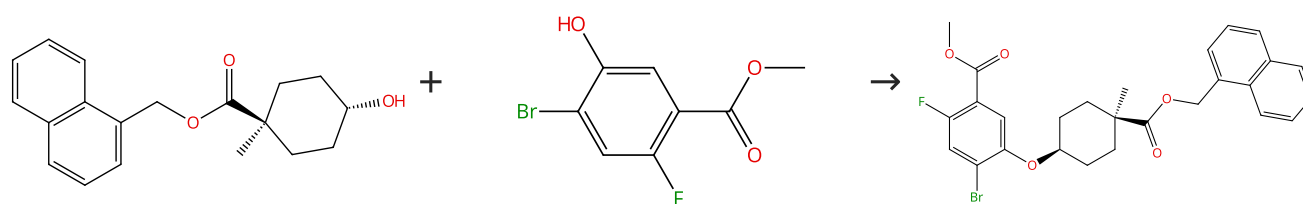
1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate

Solvents: Chloroform; rt; 0 °C; 3 h, rt

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Suppliers (35)

Relative stereochemistry shown

Suppliers (7)

31-614-CAS-40332111

Steps: 1 Yield: 100%

Discovery of Clinical Candidate AZD5462, a Selective Oral Allosteric RFXP1 Agonist for Treatment of Heart Failure

By: Granberg, Kenneth L.; et al

Journal of Medicinal Chemistry (2024), 67(6), 4419-4441.

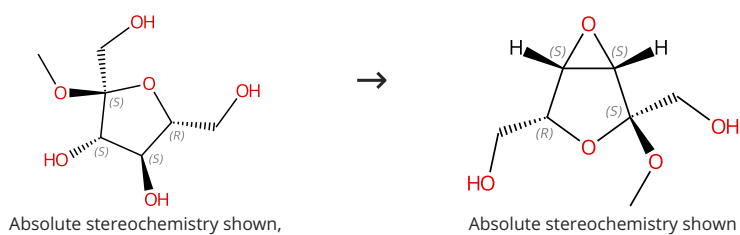
1.1 Reagents: Triphenylphosphine, 1,2-Bis(2-methoxyethyl) 1,2-diazenedicarboxylate

Solvents: Tetrahydrofuran; 40 °C; 4 h, rt

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (14)

Suppliers (2)

31-614-CAS-36851581

Steps: 1 Yield: 100%

Diphenoxytriphenylphosphorane: A potent carbohydrate cyclodehydration reagent1.1 **Reagents:** Phenol, Triphenylphosphine, Diisopropyl azodicarboxylate**Solvents:** Tetrahydrofuran; 15 min, 0 °C; 30 min, 0 °C

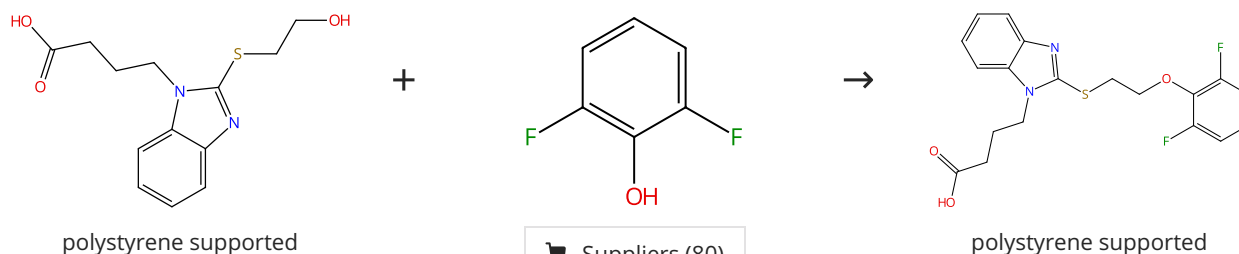
By: Sunde-Brown, Peter; et al

Tetrahedron (2022), 120, 132877.

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

31-012-CAS-16504587

Steps: 1 Yield: 100%

Solid phase syntheses of S,N-substituted 2-mercaptopbenzoimidazoles1.1 **Reagents:** Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Chloroform; rt; 0 °C; 3 h, rt

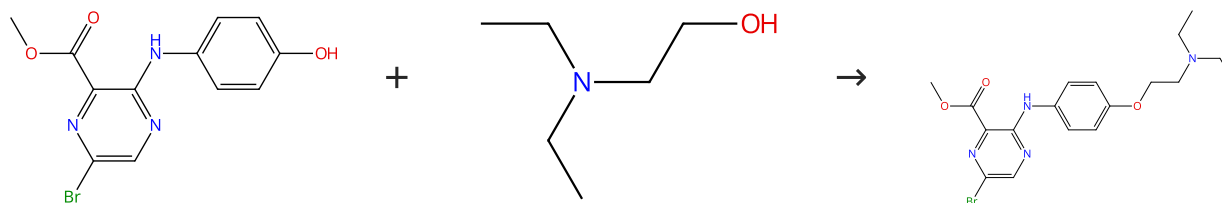
By: Susanto, Steven; et al

RSC Advances (2016), 6(46), 39573-39576.

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (48)

31-012-CAS-1021048

Steps: 1 Yield: 100%

Efficient Three-Step One-Pot Synthesis of a Novel 2,3,5-Substituted Pyrazine Library1.1 **Reagents:** Triphenylphosphine (resin-bound), Di-*tert*-butyl azodicarboxylate**Solvents:** Dichloromethane; 1 h, 0 °C; 16 h, rt

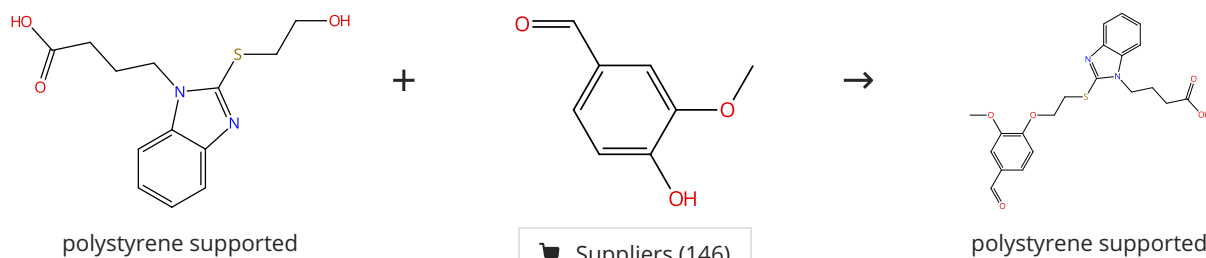
By: Delvare, Christian; et al

ACS Combinatorial Science (2011), 13(5), 449-452.

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-16504588

Steps: 1 Yield: 100%

Solid phase syntheses of *S,N*-substituted 2-mercaptobenzoimidazoles

1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Chloroform; rt; 0 °C; 3 h, rt

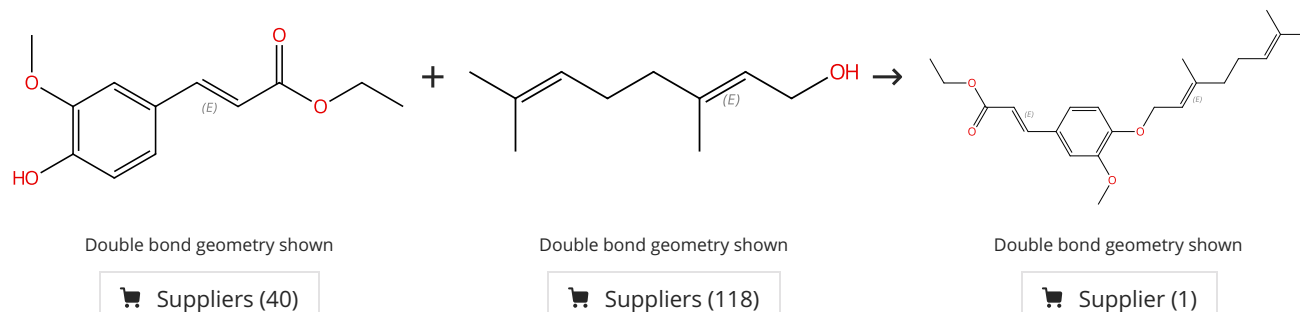
By: Susanto, Steven; et al

RSC Advances (2016), 6(46), 39573-39576.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-11222352

Steps: 1 Yield: 100%

Facile preparation of ethyl 3-(4-geranyloxy-3-methoxyphenyl)-2-propenoate (EGMP) and related compounds for the practical use of EGMP as a cancer chemopreventive agent

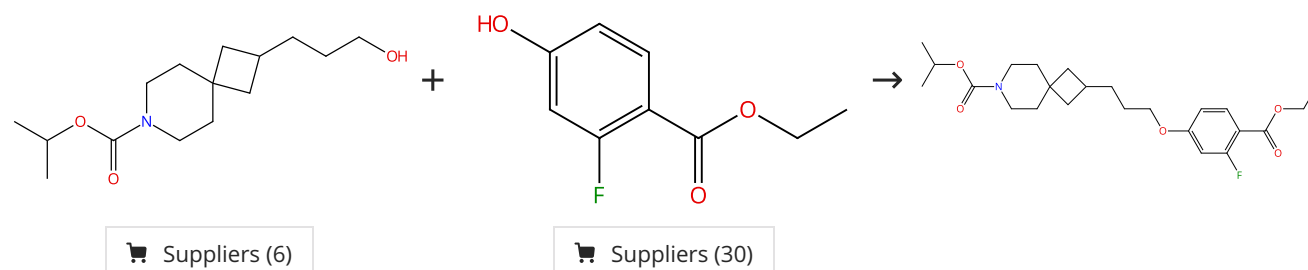
1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Toluene

By: Hosoda, Asao; et al

ITE Letters on Batteries, New Technologies & Medicine (2001), 2(5), 659-662.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-19139266

Steps: 1 Yield: 100%

Design, synthesis and biological evaluation of novel 7-azaspiro[3.5]nonane derivatives as GPR119 agonists

1.1 Reagents: Tributylphosphine, *N,N,N',N'*-Tetramethylazodicarboxamide

By: Matsuda, Daisuke; et al

Bioorganic & Medicinal Chemistry (2018), 26(8), 1832-1847.

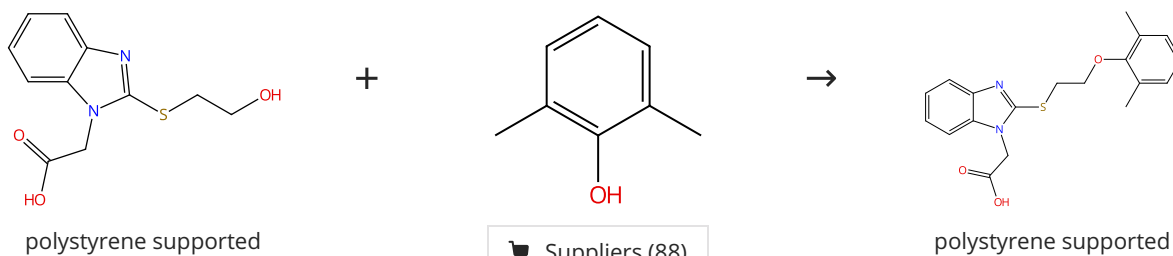
Solvents: Tetrahydrofuran; 2 h, 60 °C

1.2 Reagents: Water

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-16504585

Steps: 1 Yield: 100%

Solid phase syntheses of S,N-substituted 2-mercaptobenzimidazoles

1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
 Solvents: Chloroform; rt; 0 °C; 3 h, rt

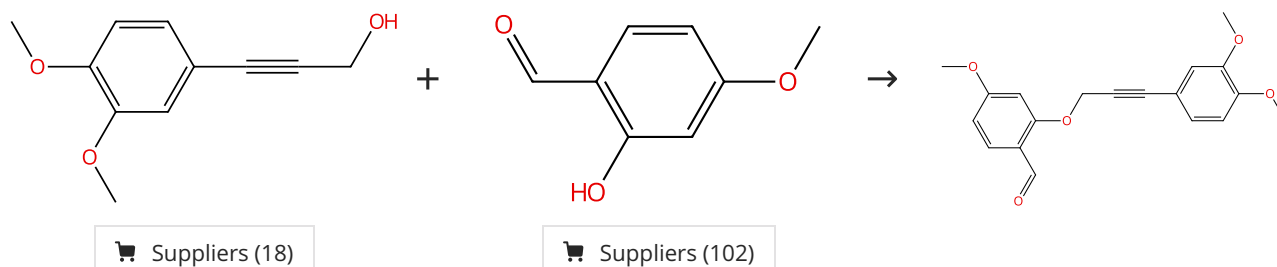
By: Susanto, Steven; et al

RSC Advances (2016), 6(46), 39573-39576.

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-7981353

Steps: 1 Yield: 100%

Total Synthesis of Brazilin

1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
 Solvents: Dichloromethane; 0 °C; 18 h, rt

By: Jung, Youngeun; et al

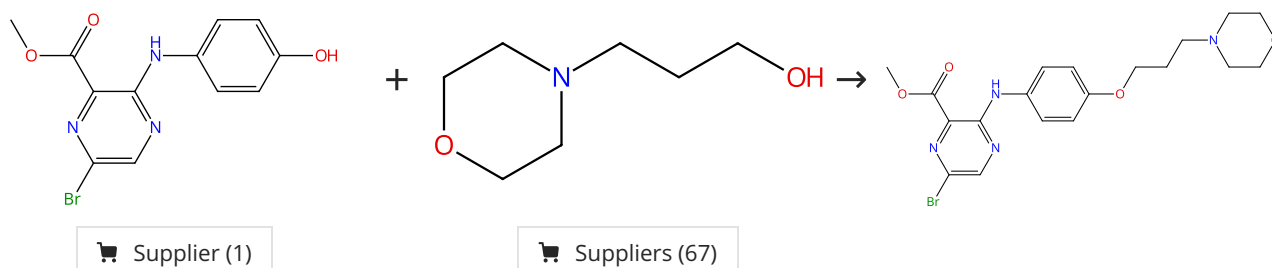
Journal of Organic Chemistry (2015), 80(3), 2001-2005.

1.2 Solvents: Water; rt

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-10150222

Steps: 1 Yield: 100%

Efficient Three-Step One-Pot Synthesis of a Novel 2,3,5-Substituted Pyrazine Library

1.1 Reagents: Triphenylphosphine (resin-bound), Di-*tert*-butyl azodicarboxylate

By: Delvare, Christian; et al

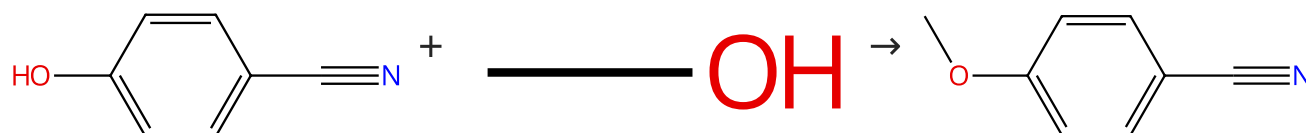
ACS Combinatorial Science (2011), 13(5), 449-452.

Solvents: Dichloromethane; 1 h, 0 °C; 16 h, rt

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

Suppliers (444)

Suppliers (89)

31-012-CAS-3128019

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Triphenylphosphine, 1,2-Bis(2-tricyclo[3.3.1.1^{3,7}]dec-1-ylethyl) 1,2-diazenedicarboxylate
Solvents: Tetrahydrofuran; overnight, rt

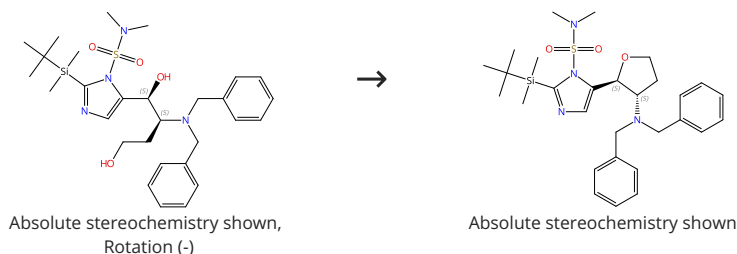
Separation tagging with cyclodextrin-binding groups:
 Mitsunobu reactions with bis(2-(1-adamantyl)ethyl) azodicarboxylate (BadEAD) and bis(1-adamantylmethyl) azodicarboxylate (BadMAD)

By: Dandapani, Sivaraman; et al

Tetrahedron Letters (2004), 45(35), 6653-6656.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-11938043

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Tributylphosphine, *N,N,N',N'*-Tetramethylazodicarboxamide
Solvents: Benzene
 1.2 **Solvents:** Ethyl acetate

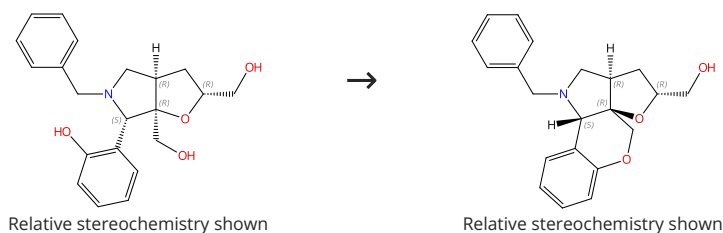
Stereoselective synthesis of 4(5)-[(2*S*,3*S*)- and (2*R*,3*R*)-3-aminotetrahydrofuran-2-yl]imidazoles using modified and standard Mitsunobu cyclizations: synthetic studies toward novel histamine H₃-ligands

By: Harusawa, Shinya; et al

Heterocycles (2000), 53(12), 2739-2751.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



31-012-CAS-15656394

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Triphenylphosphine, Di-*tert*-butyl azodicarboxylate
Solvents: Dichloromethane; 10 °C; 30 min, 10 °C

Synthesis of sp³-rich scaffolds for molecular libraries through complexity-generating cascade reactions

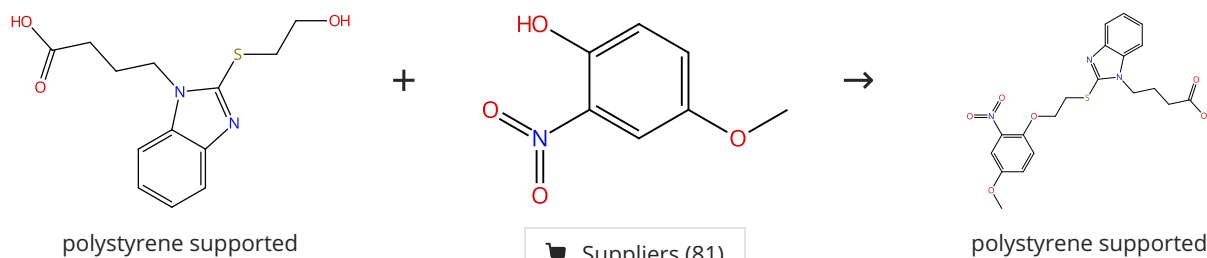
By: Flagstad, T.; et al

Organic & Biomolecular Chemistry (2016), 14(21), 4943-4946.

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (81)

31-012-CAS-16504590

Steps: 1 Yield: 100%

Solid phase syntheses of *S,N*-substituted 2-mercaptobenzoimidazoles

1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Chloroform; rt; 0 °C; 3 h, rt

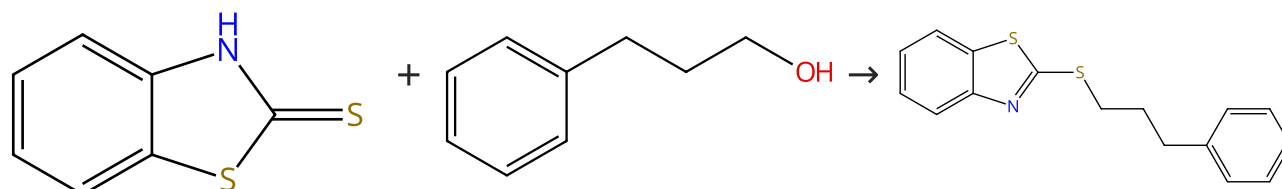
By: Susanto, Steven; et al

Experimental Protocols

RSC Advances (2016), 6(46), 39573-39576.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (111)

Suppliers (87)

Suppliers (4)

31-012-CAS-21343936

Steps: 1 Yield: 100%

Systematic Evaluation of 2-Arylazocarboxylates and 2-Arylazocarboxamides as Mitsunobu Reagents

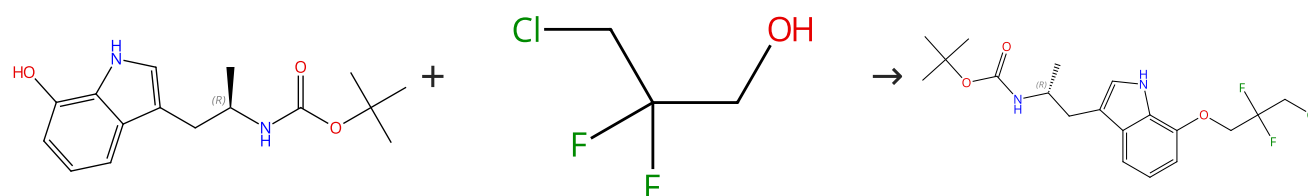
1.1 Reagents: Triphenylphosphine, Ethyl 2-(3,4-dichlorophenyl) diazenecarboxylate
Solvents: Toluene; 5 h, rt

By: Hirose, Daisuke; et al

Journal of Organic Chemistry (2018), 83(8), 4712-4729.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (4)

31-012-CAS-14166978

Steps: 1 Yield: 100%

Discovery of 1,7-cyclized indoles as a new class of potent and highly selective human β_3 -adrenergic receptor agonists with high cell permeability

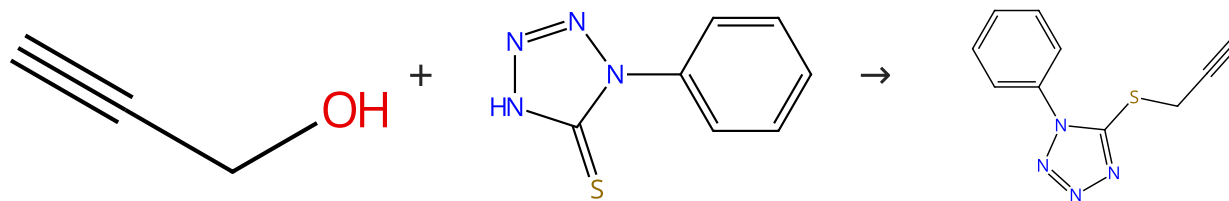
1.1 Reagents: (*R*)-2-Methyl-CBS-oxazaborolidine
Solvents: Toluene; 18 h, reflux

By: Mizuno, Kazuhiro; et al

Bioorganic & Medicinal Chemistry (2005), 13(3), 855-868.

Scheme 22 (5 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (44)

Suppliers (76)

Suppliers (12)

31-012-CAS-14236272

Steps: 1 Yield: 100%

A convenient preparation of di-p-chlorobenzyl azodicarboxylate (DCAD) for Mitsunobu couplings

By: Taft, Benjamin R.; et al

Synthesis (2009), (2), 332-334.

1.1 **Reagents:** Triphenylphosphine, 1,2-Bis[(4-chlorophenyl) methyl] 1,2-diazenedicarboxylate
Solvents: Dichloromethane; 1.5 h, rt

Experimental Protocols

31-012-CAS-12098478

Steps: 1 Yield: 100%

A convenient preparation of di-p-chlorobenzyl azodicarboxylate (DCAD) for Mitsunobu couplings

By: Taft, Benjamin R.; et al

Synthesis (2009), (2), 332-334.

1.1 **Reagents:** Triphenylphosphine, Diethyl azodicarboxylate
Solvents: Dichloromethane; 1.5 h, rt

Experimental Protocols

31-012-CAS-8342944

Steps: 1 Yield: 100%

Simplification of the Mitsunobu reaction. Di-p-chlorobenzyl azodicarboxylate: A new azodicarboxylate

By: Lipshutz, Bruce H.; et al

Organic Letters (2006), 8(22), 5069-5072.

1.1 **Reagents:** Triphenylphosphine, Diethyl azodicarboxylate
Solvents: Dichloromethane; 1.5 h, rt

Experimental Protocols

31-012-CAS-10474475

Steps: 1 Yield: 100%

Simplification of the Mitsunobu reaction. Di-p-chlorobenzyl azodicarboxylate: A new azodicarboxylate

By: Lipshutz, Bruce H.; et al

Organic Letters (2006), 8(22), 5069-5072.

1.1 **Reagents:** Triphenylphosphine, 1,2-Bis[(4-chlorophenyl) methyl] 1,2-diazenedicarboxylate
Solvents: Dichloromethane; 1.5 h, rt

Experimental Protocols

31-012-CAS-18276818

Steps: 1 Yield: 99%

Nonsymmetrical Azocarbonamide Carboxylates as Effective Mitsunobu Reagents

By: Furkert, Daniel P.; et al

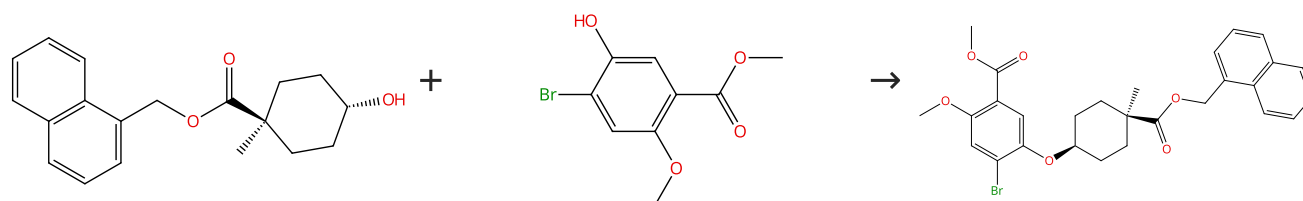
European Journal of Organic Chemistry (2014), 2014(35), 7806-7809.

1.1 **Reagents:** Triphenylphosphine, Phenylmethyl 2-[(dimethylamino)carbonyl]diazene-carboxylate
Solvents: Toluene; rt; 2 h, rt

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

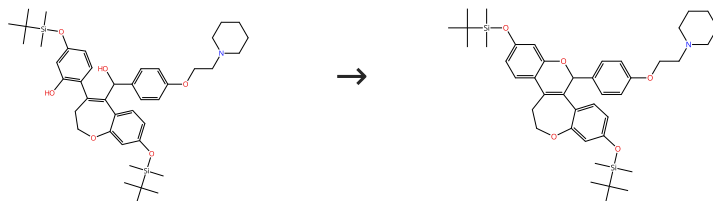
Suppliers (7)

Suppliers (13)

31-614-CAS-40331989	Steps: 1 Yield: 100%	Discovery of Clinical Candidate AZD5462, a Selective Oral Allosteric RFXP1 Agonist for Treatment of Heart Failure By: Granberg, Kenneth L.; et al Journal of Medicinal Chemistry (2024), 67(6), 4419-4441.
1.1 Reagents: Triphenylphosphine, 1,2-Bis(2-methoxyethyl) 1,2-diazenedicarboxylate Solvents: Tetrahydrofuran; 1 h, 40 °C		
Experimental Protocols		

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%

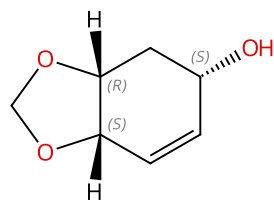


Suppliers (2)

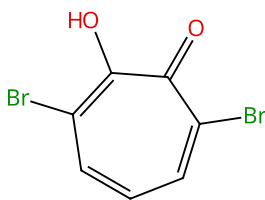
31-012-CAS-8188664	Steps: 1 Yield: 99%	Synthesis of Tetracyclic Heterocompounds as Selective Estrogen Receptor Modulators. Part 2. Process Improvement for Scale-Up Of 2,5,8-Substituted 11,12-Dihydro-5H-6,13-dioxabenz[3,4]cyclohepta-[1,2-a]naphthalene Derivatives By: Li, Xun; et al Organic Process Research & Development (2007), 11(4), 731-738.
1.1 Reagents: Hydrochloric acid Solvents: Dichloromethane, Water; 15 min, 10 °C; 10 °C → 20 °C; 30 min, 20 °C		
Experimental Protocols		

Scheme 25 (1 Reaction)

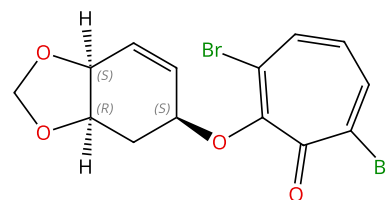
Steps: 1 Yield: 99%



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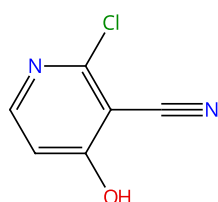


Suppliers (7)

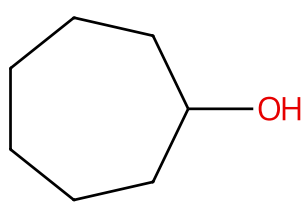
31-012-CAS-15217943	Steps: 1 Yield: 99%	Tropolonyl ethers of saccharides and cyclitol derivatives By: Houe, Henri; et al Tetrahedron Letters (2002), 43(50), 9217-9220.
1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate Solvents: Diethyl ether; rt; 1 h, rt		
1.2 Solvents: Dichloromethane		
1.3 Reagents: Hydrochloric acid Solvents: Water		

Scheme 26 (1 Reaction)

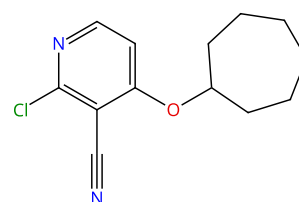
Steps: 1 Yield: 99%



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Suppliers (28)

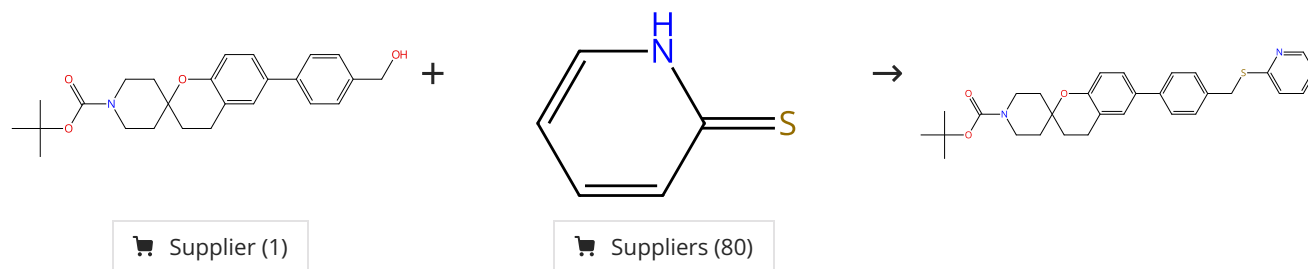
Suppliers (66)

Suppliers (2)

31-012-CAS-16714695 Steps: 1 Yield: 99% 1.1 Reagents: Triphenylphosphine, Diethyl azodicarboxylate Solvents: Toluene, Tetrahydrofuran; 0 °C; 20 h, 0 °C	An Efficient Method For The Preparation Of 4-Alkoxy-Substituted Thieno[2,3-b]Pyridines By: Saito, Keiji; et al Heterocycles (2014), 89(6), 1491-1501.
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Scheme 27 (1 Reaction)

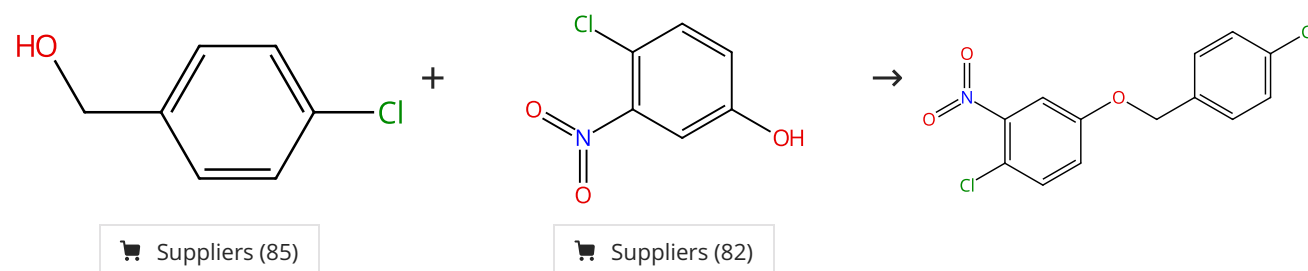
Steps: 1 Yield: 99%



31-012-CAS-18069684 Steps: 1 Yield: 99% 1.1 Reagents: Tributylphosphine, Diisopropyl azodicarboxylate Solvents: Tetrahydrofuran	A facile method for converting alcohol to thioether and its application in the synthesis of a novel GPR119 agonist By: Koshizawa, Tomoaki; et al Tetrahedron (2015), 71(21), 3231-3236.
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Scheme 28 (1 Reaction)

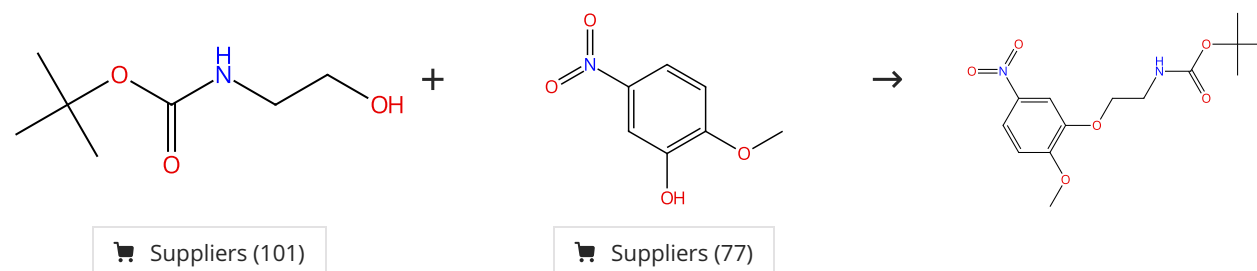
Steps: 1 Yield: 99%



31-012-CAS-8393191 Steps: 1 Yield: 99% 1.1 Reagents: 1,2-Bis(3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluorooctyl) 1, 2-diazenedicarboxylate, [4-(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-Heptafluorodecyl)phenyl]diphenylphosphine Solvents: Tetrahydrofuran; 1 h, rt Experimental Protocols	96-Well Plate-to-Plate Gravity Fluorous Solid-Phase Extraction (F-SPE) for Solution-Phase Library Purification By: Zhang, Wei; et al Journal of Combinatorial Chemistry (2007), 9(5), 836-843.
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Scheme 29 (1 Reaction)

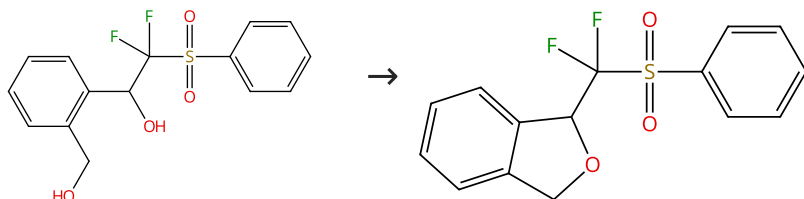
Steps: 1 Yield: 99%



31-012-CAS-19158705	Steps: 1 Yield: 99%	Structure-activity relationship investigation of Phe-Arg mimetic region of human glutamyl cyclase inhibitors
1.1 Reagents: Triphenylphosphine, Diethyl azodicarboxylate Solvents: Dichloromethane; 30 min, rt		By: Ngo, Van T. H.; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2018), 26(12), 3133-3144.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



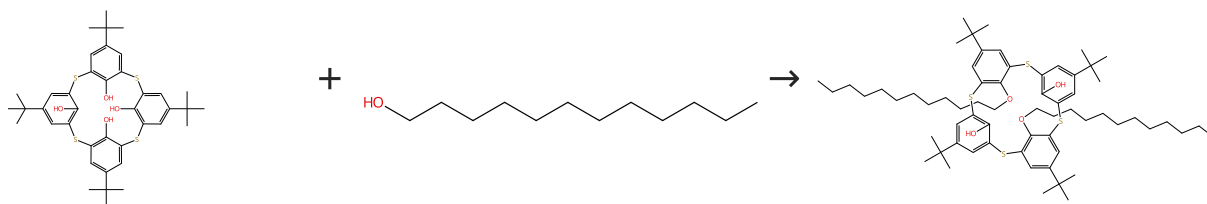
Supplier (1)

Supplier (1)

31-012-CAS-13409856	Steps: 1 Yield: 99%	Cyclodehydration of fluorinated diols using the Mitsunobu reaction: highly efficient synthesis of trifluoromethylated cyclic ethers
1.1 Reagents: Tributylphosphine, <i>N,N,N',N'</i> -Tetramethylazodicarbonyl diamide Solvents: Benzene; 12 h, rt		By: Prakash, G. K. Surya; et al
Experimental Protocols		Synthesis (2010), (11), 1891-1898.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



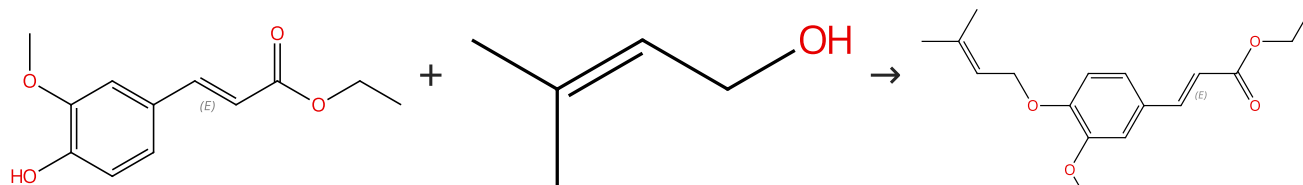
Suppliers (45)

Suppliers (99)

31-614-CAS-31585006	Steps: 1 Yield: 99%	Thiacalixarenes with Sulfur Functionalities at Lower Rim: Heavy Metal Ion Binding in Solution and 2D-Confined Space
1.1 Reagents: Triphenylphosphine, Diethyl azodicarboxylate Solvents: Toluene; rt; 8 h, 40 °C		By: Muravev, Anton; et al
Experimental Protocols		International Journal of Molecular Sciences (2022), 23(4), 2341.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (40)

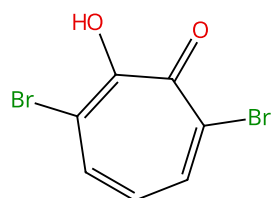
Suppliers (77)

Double bond geometry shown

31-012-CAS-15479545	Steps: 1 Yield: 99%	Facile preparation of ethyl 3-(4-geranyloxy-3-methoxyphenyl)-2-propenoate (EGMP) and related compounds for the practical use of EGMP as a cancer chemopreventive agent By: Hosoda, Asao; et al ITE Letters on Batteries, New Technologies & Medicine (2001), 2(5), 659-662.
1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate Solvents: Toluene		

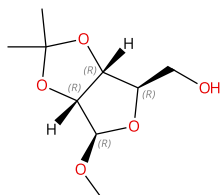
Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (7)

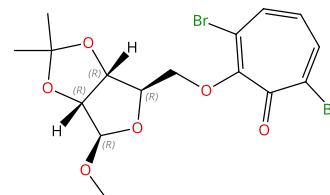
+



Absolute stereochemistry shown,
Rotation (-)

Suppliers (78)

→

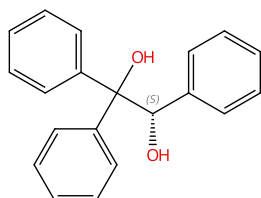


Absolute stereochemistry shown

31-012-CAS-11248367	Steps: 1 Yield: 99%	Tropolonyl ethers of saccharides and cyclitol derivatives By: Houe, Henri; et al Tetrahedron Letters (2002), 43(50), 9217-9220.
1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate Solvents: Diethyl ether; rt; 1 h, rt		
1.2 Solvents: Dichloromethane		
1.3 Reagents: Hydrochloric acid Solvents: Water		

Scheme 34 (1 Reaction)

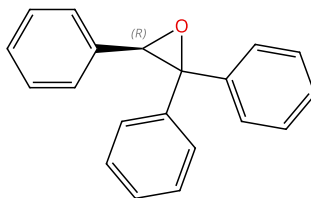
Steps: 1 Yield: 99%



Absolute stereochemistry shown,
Rotation (-)

Suppliers (69)

→



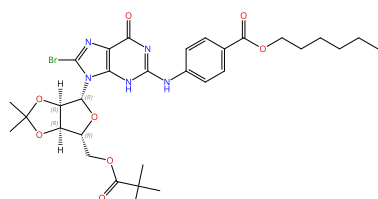
Absolute stereochemistry shown,
Rotation (-)

Suppliers (3)

31-012-CAS-1054135	Steps: 1 Yield: 99%	Phosphine-Dependent Stereoselectivity in the Mitsunobu Cyclodehydration of 1,2-Diols: Stereodivergent Approach to Triaryl-Substituted Epoxides By: Garcia-Delgado, Noemi; et al Organic Letters (2007), 9(4), 635-638.
1.1 Reagents: Diisopropyl azodicarboxylate, Trioctylphosphine Solvents: Dichloromethane; rt		
Experimental Protocols		

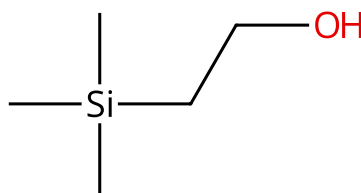
Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



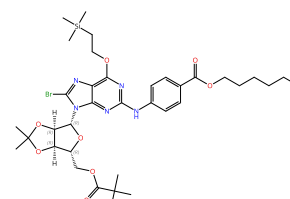
Absolute stereochemistry shown

+



Suppliers (85)

→

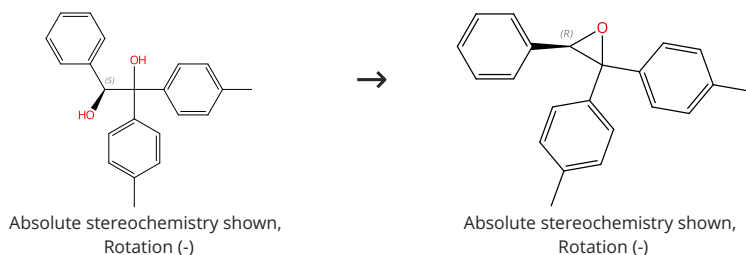


Absolute stereochemistry shown

31-012-CAS-18238781	Steps: 1 Yield: 99%	G-Arylated Hydrogen-Bonded Cyclic Tetramer Assemblies with Remarkable Thermodynamic and Kinetic Stability
1.1 Reagents: Triphenylphosphine Solvents: 1,4-Dioxane; 50 °C		By: Romero-Perez, Sonia; et al
1.2 Reagents: Diisopropyl azodicarboxylate		Organic Letters (2015), 17(11), 2664-2667.
Experimental Protocols		

Scheme 36 (1 Reaction)

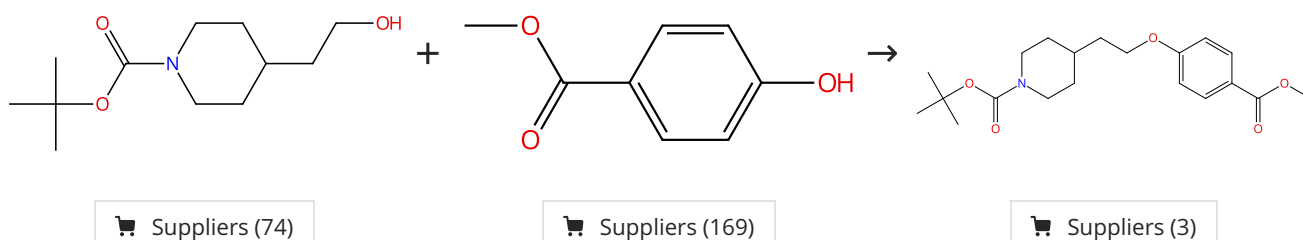
Steps: 1 Yield: 99%



31-012-CAS-14448866	Steps: 1 Yield: 99%	Phosphine-Dependent Stereoselectivity in the Mitsunobu Cyclodehydration of 1,2-Diols: Stereodivergent Approach to Triaryl-Substituted Epoxides
1.1 Reagents: Diisopropyl azodicarboxylate, Trioctylphosphine Solvents: Dichloromethane; rt		By: Garcia-Delgado, Noemi; et al
Experimental Protocols		Organic Letters (2007), 9(4), 635-638.

Scheme 37 (1 Reaction)

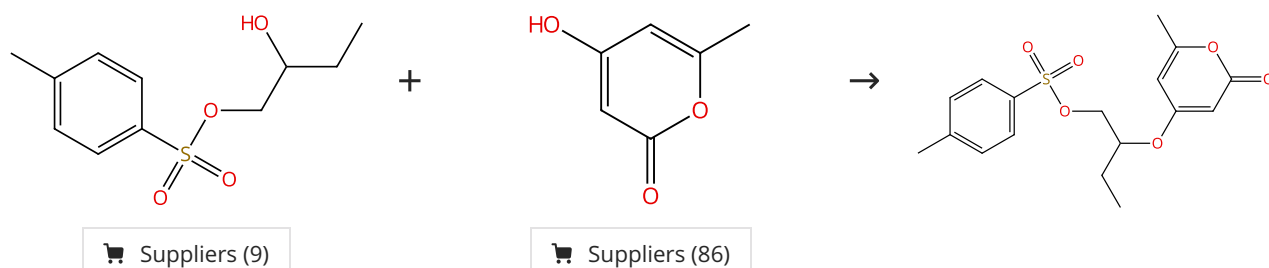
Steps: 1 Yield: 99%



31-012-CAS-9951813	Steps: 1 Yield: 99%	Synthesis of Arrays using Low Molecular Weight M PEG-Assisted Mitsunobu Reaction
1.1 Reagents: Poly(oxy-1,2-ethanediyl), α-[4-(diphenylphosphino) phenyl]-ω-methoxy-, Poly(oxy-1,2-ethanediyl), α-[[2-(ethoxycarbonyl)diazonyl]carbonyl]-ω-methoxy- Solvents: Tetrahydrofuran; 20 h, rt		By: Figlus, Marek; et al
Experimental Protocols		ACS Combinatorial Science (2011), 13(3), 280-285.

Scheme 38 (1 Reaction)

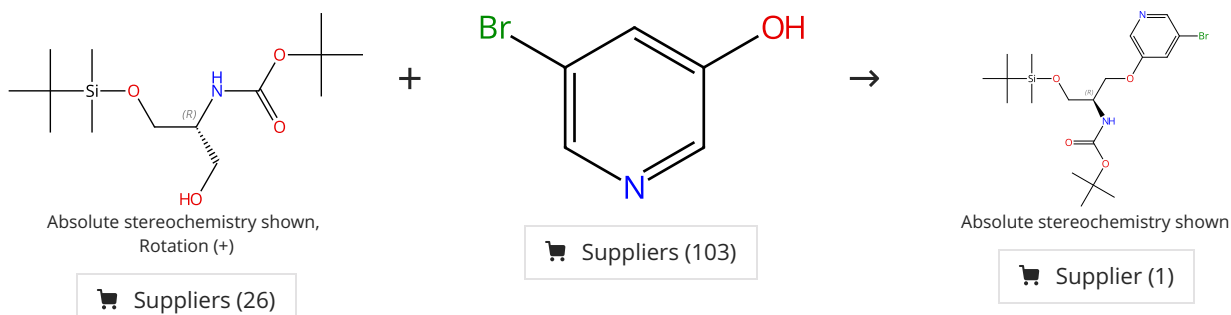
Steps: 1 Yield: 99%



31-012-CAS-15008688 Steps: 1 Yield: 99% 1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate Solvents: Dichloromethane; 10 - 30 min, 0 °C; 16 h, 18 - 25 °C	4-Hydroxy-6-alkyl-2-pyrones as nucleophilic coupling partners in Mitsunobu reactions and oxa-Michael additions By: Burns, Michael J.; et al Beilstein Journal of Organic Chemistry (2014), 10, 1159-1165, 7 pp..
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Scheme 39 (1 Reaction)

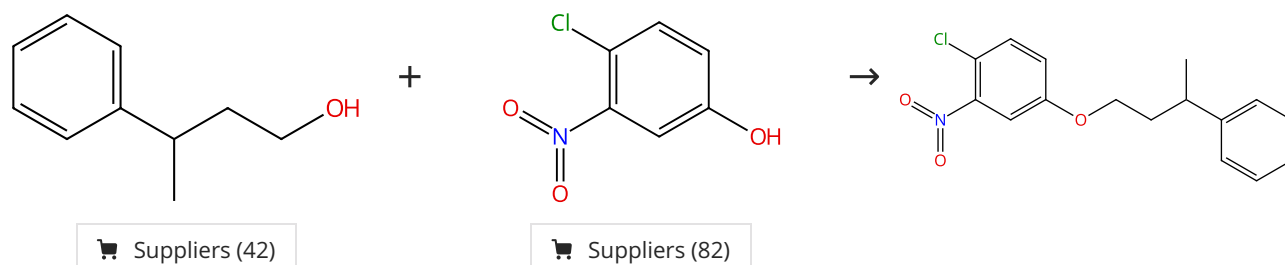
Steps: 1 Yield: 99%



31-012-CAS-3471317 Steps: 1 Yield: 99% 1.1 Reagents: Triphenylphosphine Solvents: Tetrahydrofuran; 10 min, 0 °C 1.2 Reagents: Diethyl azodicarboxylate; 0.5 h, 0 °C; 2 h, rt Experimental Protocols	Syntheses of Potent, Selective, and Orally Bioavailable Indazole-Pyridine Series of Protein Kinase B/Akt Inhibitors with Reduced Hypotension By: Zhu, Gui-Dong; et al Journal of Medicinal Chemistry (2007), 50(13), 2990-3003.
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Scheme 40 (1 Reaction)

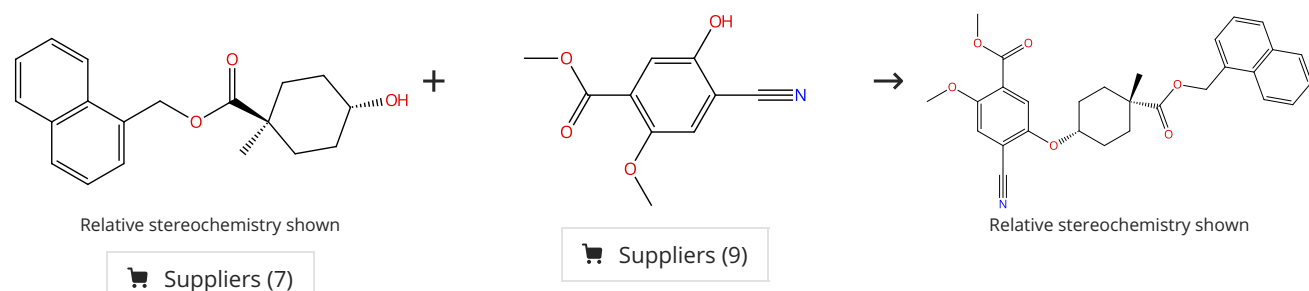
Steps: 1 Yield: 99%



31-012-CAS-4140505 Steps: 1 Yield: 99% 1.1 Reagents: 1,2-Bis(3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluorooctyl) 1,2-diazenedicarboxylate, [4-(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-Heptafluorodecyl)phenyl]diphenylphosphine Solvents: Tetrahydrofuran; 1 h, rt Experimental Protocols	96-Well Plate-to-Plate Gravity Fluorous Solid-Phase Extraction (F-SPE) for Solution-Phase Library Purification By: Zhang, Wei; et al Journal of Combinatorial Chemistry (2007), 9(5), 836-843.
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Scheme 41 (1 Reaction)

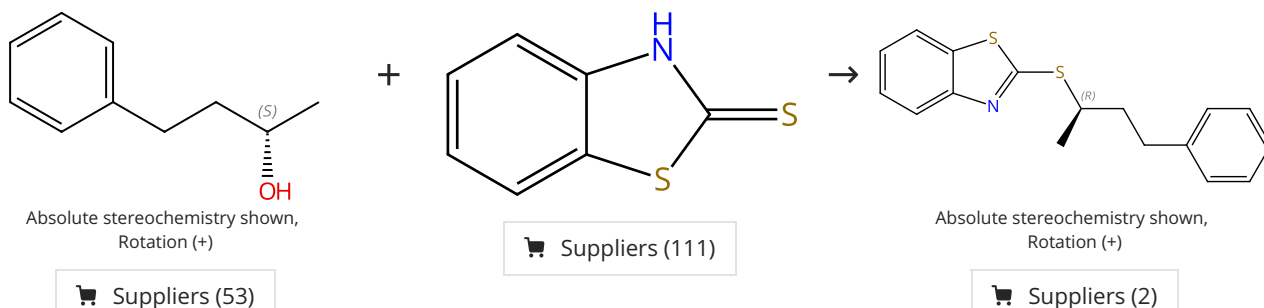
Steps: 1 Yield: 99%



31-614-CAS-40332035	Steps: 1 Yield: 99%	Discovery of Clinical Candidate AZD5462, a Selective Oral Allosteric RFXP1 Agonist for Treatment of Heart Failure
1.1 Reagents: Triphenylphosphine Solvents: Tetrahydrofuran; 10 min, 60 °C		By: Granberg, Kenneth L.; et al
1.2 Reagents: Diisopropyl azodicarboxylate; 60 °C; 14 h, 60 °C		Journal of Medicinal Chemistry (2024), 67(6), 4419-4441.
Experimental Protocols		

Scheme 42 (1 Reaction)

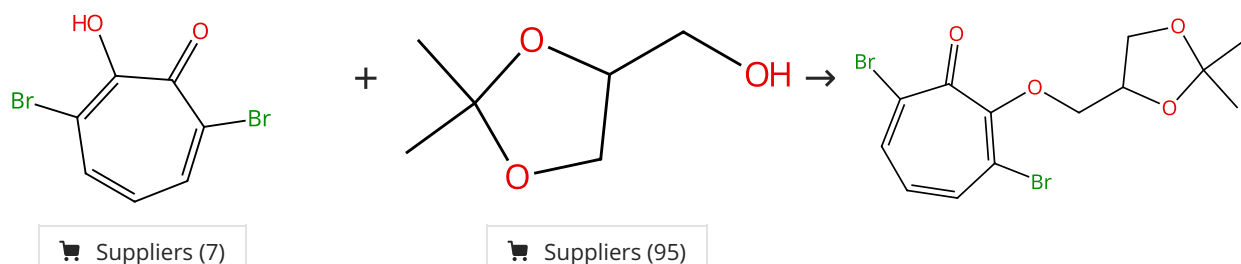
Steps: 1 Yield: 99%



31-012-CAS-2361022	Steps: 1 Yield: 99%	A new type of oxidation-reduction condensation by the combined use of phenyl diphenylphosphinite and oxidant
1.1 Reagents: Ethyl azidoacetate, Phenyl <i>P,P</i> -diphenylphosphinite Solvents: Toluene; 24 h, 40 °C		By: Mukaiyama, Teruaki; et al
		Heterocycles (2010), 80(1), 63-82.

Scheme 43 (1 Reaction)

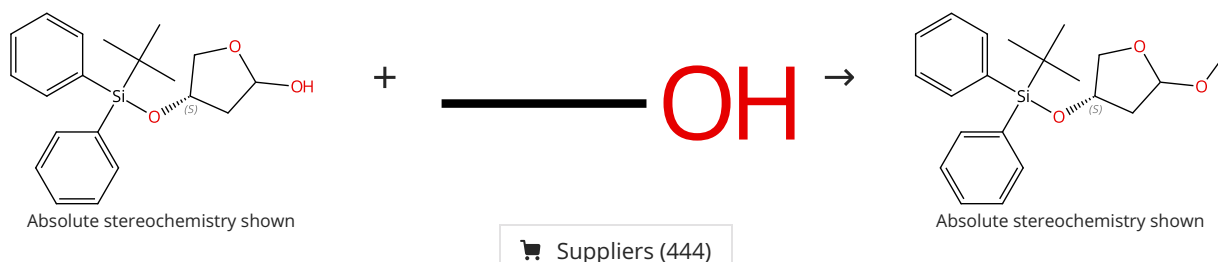
Steps: 1 Yield: 98%



31-012-CAS-2769310	Steps: 1 Yield: 98%	Tropolonyl ethers of saccharides and cyclitol derivatives
1.1 Reagents: Triphenylphosphine, Diisopropyl azodicarboxylate Solvents: Diethyl ether; rt; 1 h, rt		By: Houe, Henri; et al
1.2 Solvents: Dichloromethane		Tetrahedron Letters (2002), 43(50), 9217-9220.
1.3 Reagents: Hydrochloric acid Solvents: Water		

Scheme 44 (1 Reaction)

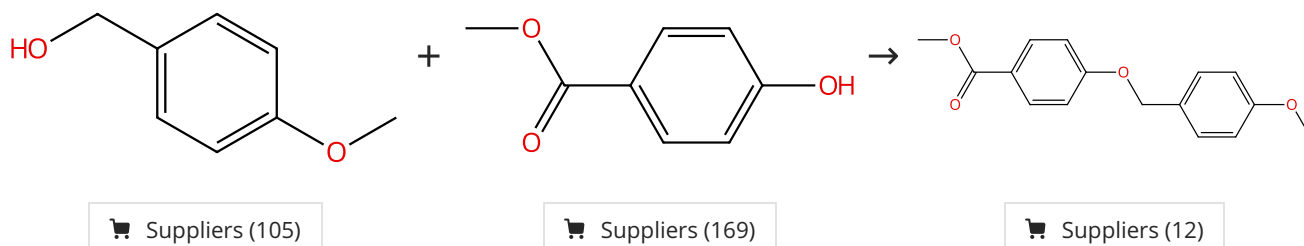
Steps: 1 Yield: 98%



31-012-CAS-8253777 Steps: 1 Yield: 98% 1.1 Reagents: Hydrochloric acid; 2 h, rt 1.2 Reagents: Triethylamine; neutralized Experimental Protocols	Synthesis of 3'-S-Phosphonomethyl-Modified Nucleoside Phosphonates with a 3'-Deoxy-3'-thio-α-L-threosyl Sugar Moiety By: Huang, Qiuya; et al European Journal of Organic Chemistry (2011), 2011(19), 3450-3457, S3450/1-S3450/23.
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Scheme 45 (1 Reaction)

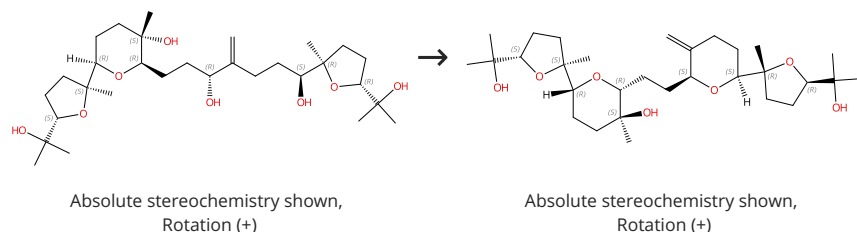
Steps: 1 Yield: 98%



31-012-CAS-8059109 Steps: 1 Yield: 98% 1.1 Reagents: 1,2-Bis(3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluorooctyl) 1, 2-diazenedicarboxylate, [4-(3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-Heptadecafluorodecyl)phenyl]diphenylphosphine Solvents: Tetrahydrofuran; 1 h, rt Experimental Protocols	96-Well Plate-to-Plate Gravity Fluorous Solid-Phase Extraction (F-SPE) for Solution-Phase Library Purification By: Zhang, Wei; et al Journal of Combinatorial Chemistry (2007), 9(5), 836-843.
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Scheme 46 (1 Reaction)

Steps: 1 Yield: 98%



31-012-CAS-14162731 Steps: 1 Yield: 98% 1.1 Solvents: Acetic anhydride, Pyridine; 36 h, 50 °C Experimental Protocols	Total Synthesis of Pseudodehydrothysiferol By: Hioki, Hideaki; et al Organic Letters (2009), 11(3), 579-582.
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